

Efficient Multimodal Quadrupedal Autonomy: Algorithm, System and Application

Abhijeet M. Kulkarni[‡], Chinmay Burgul[‡], Yuze Du, and Guoquan Huang

Abstract—While legged locomotion has achieved notable progress, with striking demonstrations showcasing its potential, quadrupedal robots continue to face challenges in realizing efficient and reliable autonomy. This limitation remains a significant obstacle to their widespread use in high-risk, real-world scenarios. In this paper, we take a critical first step toward achieving *quadrupedal autonomy* – extending beyond locomotion alone – through a modular design that seamlessly integrates multimodal state estimation, online mapping (perception), and ESDF-based planning. We incorporate these quadrupedal-tailored algorithms into our *Ghost Vision 60* quadrupedal platform (see Fig. 1) to develop an efficient autonomy stack, which is validated by an autonomous exploration application in a GPS-denied indoor environment.

I. INTRODUCTION

Quadrupedal robots, given their capacity to traverse difficult terrains that wheeled systems cannot, are ideally suited for applications such as disaster response, planetary exploration, and industrial inspection. For example, Fig. 1 depicts our quadruped robot (*Ghost Vision 60*), highlighting its capacity for deployment in real world to undertake tasks including environmental monitoring and the collection of scientific data. Nevertheless, despite notable recent developments [1], these robots have not yet attained the efficient and robust *autonomy* essential for reliable operation within high-risk domains. Presently, sensing and perception systems such as cameras encounter substantial limitations in environments marked by low light, or minimal texture, thereby compromising the accuracy and efficacy of state estimation and motion planning. These impediments are exacerbated by the stringent resource constraints imposed upon *any* real robot during field operations, which pose significant challenges to effective autonomy algorithms. As such, in this paper, we take a deliberate step forward to develop efficient *quadrupedal autonomy*.

Recently, we have witnessed significant advancements on quadrupedal locomotion [1], [2]. However, these robots still lack the robust and assured autonomy needed for reliable performance in the wild. In particular, the lack of efficient state estimation and motion planning algorithms tailored to quadrupedal legged systems severely limits their autonomy. Many existing approaches (e.g., [3]) rely on computationally expensive trajectory optimization or reinforcement learning methods that, while effective in controlled settings, often struggle with real-time performance in unknown environments. State estimation typically prioritizes accuracy over



Fig. 1: A *Ghost Vision 60* quadruped robot equipped with multimodal sensors such as Lidar and cameras serves as the hardware platform for the proposed autonomy algorithms.

efficiency and robustness, failing to fully exploit the inherent constraints of quadrupedal locomotion [4]–[8]. As a result, it remains slow and unreliable, particularly when optimally fusing multimodal data. Similarly, motion planning for quadrupeds is largely adapted from wheeled and aerial robots [9]–[11]. Many planners rely on precomputed trajectories or sampling-based methods that are computationally intensive and slow to react to environmental changes. This inefficiency limits the ability of quadrupedal robots to navigate unknowns autonomously.

In this paper, we take a pivotal first step toward achieving quadrupedal autonomy—beyond mere locomotion—by adopting a modular design that seamlessly integrates multimodal state estimation, online multimodal mapping (perception), and Euclidean Signed Distance Function (ESDF)-based quadrupedal planning, which would equip quadrupeds with efficient autonomy to tackle diverse tasks such as autonomous exploration. Key innovations include a lightweight, kinematics-constrained, gait-informed visual-inertial estimation module that offers accurate navigation even in GPS-denied environments. Based on the precise localization, an online multimodal ESDF-based mapper is developed by extending the off-the-shelf *nvblox* library [12] to efficiently fuse LiDAR and RGBD data available to the quadrupedal robot. Furthermore, we build a resource-aware planner that leverages the ESDF map directly and incorporates quadrupedal kinodynamic constraints, allowing to dynamically adapt to

The authors are with the Robot Perception and Navigation Group (RPNP), University of Delaware, Newark, DE 19716, USA. {amkulk, cmburgul, yuzedu, ghuang}@udel.edu

[‡]The authors made equal contributions to this work.

its environment despite limited computing resources. As a culminating point, we integrate our quadrupedal-tailored autonomy algorithms – spanning state estimation, mapping/perception, and motion planning – into the *Ghost Vision 60* platform to build an efficient and robust quadrupedal autonomy stack. To validate this system, we develop an autonomous exploration application and carry out preliminary testing in a GPS-denied indoor environment.

II. RELATED WORK

As quadrupedal robots continue to advance, a substantial body of literature has emerged (e.g., [13]). In this section, we provide a brief review of the work most relevant to the proposed method, without intention to present a comprehensive survey on quadrupedal robotics.

A. Perception and Estimation

Truncated Signed Distance Function (TSDF) [14] is often used to fuse range data into a coherent volumetric representation. In particular, voxblox [15] has shown the viability of incremental TSDF and ESDF computation for real-time on-board applications. GPU-accelerated extensions like nvblox [12] further enhance these methods, enabling high-resolution mapping in large-scale and dynamic environments, which will be also leveraged in this work. For legged locomotion, elevation maps are often used to model the topography of the ground and are typically generated by fusing data from sensors such as LiDAR and stereo/depth cameras [16]–[18]. Such maps are useful for navigation, obstacle avoidance, and path planning, as they capture both geometric and semantic details of the ground surface.

Efficient state estimation is needed for legged robots that are typically equipped with high-rate IMUs and leg encoders. Proprioceptive-based state estimators [8], [19], [20] perform well at lower velocities, high friction, and rigid terrain but tend to drift during foot slippage, and deformable terrains. To address this issue, one may detect unstable foot contacts and reduce their influence on estimation [21], [22]. While using exteroceptive sensors with legged robots, during the gait motion, ground contact generates impact forces that corrupts the inertial measurements and introduce disturbance into exteroceptive data. While incorporating leg kinematics within Visual Inertial Navigation System (VINS) improves estimation accuracy [6]–[8], [23]–[28], achieving robust estimation in low-light and textureless environments which remains a challenge for VINS.

B. Planning and Control

Recent model-based control is often augmented with the perception information by either sequentially or jointly planning the footholds and whole-body motion but limiting complex movements [11], [29]–[32]. Learning-based locomotion prevails lately and has resulted in some agile controllers with less manual effort than model-based controllers. Although a policy may be learned in simulations to maximize a reward that encodes the task, the hurdle of sim-to-real gap remains. Researchers have attempted to address this with domain randomization [33], [34], rapid motor adaptation [35], [36],

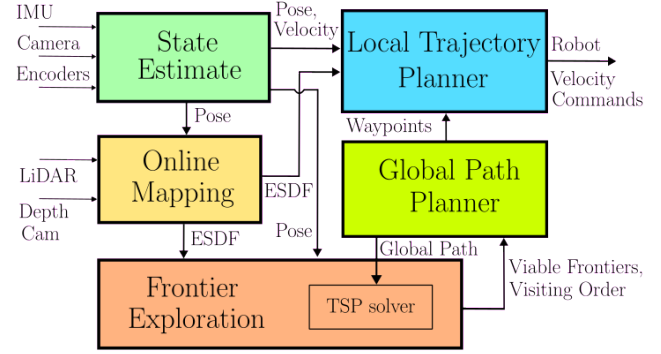


Fig. 2: System diagram of the proposed quadrupedal autonomy stack, shows all the modules and connections.

and multi-modal delay randomization [37] to induce robustness against multiple sensors. While perception-informed policies were also deployed with quadrupeds for tasks such as obstacle avoidance [38], they can be sample inefficient and unreliable being black-boxed.

III. OVERVIEW OF QUADRUPEDAL AUTONOMY

Fig. 1 shows the hardware platform of our quadrupedal robot (*Ghost Vision 60*), which is equipped with a set of multimodal sensors, including Ouster Lidar OS1 (or Velodyne VLP-16), monocular cameras and RGBD sensors (RealSense D435), along with the IMU and joint encoders. There are two embedded computers onboard: an NVIDIA Xavier, which handles low-level control, and an Orin, which manages the high-level autonomy stack. The low-latency communication between them is via Ethernet with ROS2 Data Distribution Service (DDS).

Fig. 2 illustrates the system diagram of the proposed quadrupedal autonomy stack. While not novel in itself, this baseline system facilitates the integration of several innovative modular algorithms. The State Estimation module fuses the inputs from IMU at 500 Hz, monocular Camera at 30 Hz, leg joint encoders at 250 Hz and estimates the body pose and velocity at 30 Hz. Online Mapping module takes the pointclouds of the Lidar and depth sensor along with pose estimates and generates an ESDF map for planning at 1Hz. The Frontier Exploration module takes in the ESDF map as input and generates frontier centroids using a traveling salesman problem. Global Planner generates a viable path from the frontier centroids and outputs the next waypoint goal, while the Local Trajectory planner takes this waypoint goal, optimizes a bezier curve trajectory, and outputs velocities and yaw rate sent to the low-level controller to execute.

IV. LIGHTWEIGHT MULTIMODAL STATE ESTIMATION

State estimation is at the core of the proposed quadrupedal autonomy, providing accurate real-time estimates to enable the downstream perception and planning modules. The proposed multimodal state estimator is built upon our prior work [27] and efficiently fuses the IMU readings, visual feature measurements and robot kinematics, as well as, for the first time, the legged motion gait constraints.

A. Visual-Inertial Estimation

Assuming the IMU coincides with the center of mass (COM) of the robot's body or floating base, the state vector includes the toe positions (which serve as the landmarks) and the standard navigation states:¹

$$\mathbf{x}_k = [\mathbf{x}_{I_k}^\top \quad \mathbf{x}_{C_k}^\top \quad \mathbf{x}_\tau^\top]^\top \quad (1)$$

$$\mathbf{x}_{I_k} = [{}^G \bar{q}^\top \quad {}^G \mathbf{p}_{I_k}^\top \quad {}^G \mathbf{v}_{I_k}^\top \quad \mathbf{b}_g^\top \quad \mathbf{b}_a^\top]^\top \quad (2)$$

$$\mathbf{x}_{C_k} = [{}^G \bar{q}^\top \quad {}^G \mathbf{p}_{I_{k-1}}^\top \quad \dots \quad {}^{I_{k-n}} \bar{q}^\top \quad {}^G \mathbf{p}_{I_{k-n}}^\top]^\top \quad (3)$$

$$\mathbf{x}_\tau = [{}^G \mathbf{p}_{\tau_1}^\top \quad \dots \quad {}^G \mathbf{p}_{\tau_w}^\top]^\top \quad (4)$$

where $\mathbf{x}_{I_k}$ is the current body/IMU state, $\mathbf{x}_{C_k}$ are the n historical cloned poses in the global frame $\{G\}$, $\mathbf{x}_\tau$ are the w static toe landmark positions. ${}^I_k \bar{q}$ is the JPL unit quaternion corresponding to the rotation ${}^I_k \mathbf{R}$ from $\{G\}$ to $\{I\}$, ${}^G \mathbf{p}_{I_k}$ and ${}^G \mathbf{v}_{I_k}$ are the position and velocity, and $\mathbf{b}_g$ and $\mathbf{b}_a$ are the biases of the gyroscope and accelerometer, respectively. The navigation state evolves over time with the IMU kinematics, using incoming linear acceleration ${}^I \mathbf{a}_k$, and angular velocity ${}^I \boldsymbol{\omega}_k$, which is given by [39]:

$$\mathbf{x}_{I_{k+1}} = \mathbf{f}_I(\mathbf{x}_{I_k}, {}^I \mathbf{a}_k, {}^I \boldsymbol{\omega}_k, \mathbf{n}_I) \quad (5)$$

where $\mathbf{n}_I = [\mathbf{n}_a^\top \quad \mathbf{n}_g^\top \quad \mathbf{n}_{wa}^\top \quad \mathbf{n}_{wg}^\top]^\top$ contains the zero-mean white Gaussian noises and random walks of the biases.

We detect and track static environmental features over images with optical flow, whose measurement $\mathbf{z}_{C_k}$ at timestep k is obtained by projecting the 3D global position ${}^G \mathbf{p}_f$ onto the image plane:

$$\mathbf{z}_{C_k} = \mathbf{h}_c(\mathbf{x}_{C_k}, {}^C \mathbf{p}_f) + \mathbf{n}_{C_k} \quad (6)$$

$$\text{where } {}^C \mathbf{p}_f = {}^C \mathbf{R}_G^I \mathbf{R} ({}^G \mathbf{p}_f - {}^G \mathbf{p}_{I_k}) + {}^C \mathbf{p}_I \quad (7)$$

where $\{{}^C \mathbf{R}, {}^C \mathbf{p}_I\}$ is the IMU-camera extrinsic transformation which is assumed to be known otherwise can be calibrated either online or offline [40], and $\mathbf{n}_{C_k}$ is the camera measurement noise. We perform the MSCKF-based VINS as the backbone estimator to efficiently propagate with the IMU readings and update with the visual measurements [41], [42].

B. Incorporating Kinematics Constraints

When the robot is walking, and the leg is in contact with ground, we compute the position of toe as a landmark ${}^G \mathbf{p}_\tau$ in global frame using the forward kinematics ${}^I \mathbf{p}_\tau(\boldsymbol{\theta}_m)$ and IMU pose estimates, then initialize it into the state (4). During the rest of the contact phase, we also update the state with this leg-kinematic measurement [27]:

$$\mathbf{z}_\ell = \mathbf{h}_\ell(\mathbf{x}_I, {}^G \mathbf{p}_\tau, {}^I \mathbf{p}_\tau) = {}^I \mathbf{R} ({}^G \mathbf{p}_\tau - {}^G \mathbf{p}_I) - {}^I \mathbf{p}_\tau(\boldsymbol{\theta}_m) \quad (8)$$

$$\text{where } {}^I \mathbf{p}_\tau(\boldsymbol{\theta}_m) = \mathbf{g}_\ell({}^I \mathbf{R}, {}^I \mathbf{p}_L, \ell, \boldsymbol{\theta}_m) \quad (9)$$

where $\{{}^I \mathbf{R}, {}^G \mathbf{p}_I\}$ is the IMU-Leg extrinsic transformation, ${}^G \mathbf{p}_\tau$ is toe landmark, $\boldsymbol{\theta}_m$ are the leg encoder measurements, ℓ are the link lengths of leg. The leg contact is detected statistically and when the contact is lost, we marginalize out the toe landmark from the state to bound computational cost.

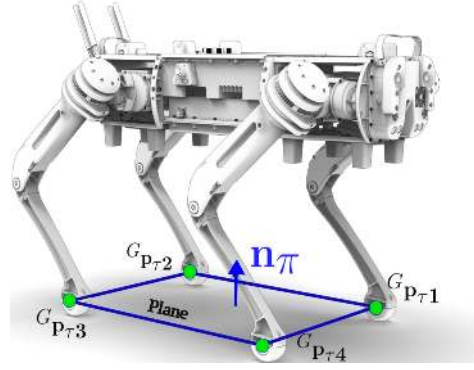


Fig. 3: Walking ground plane constructed from the toe positions in the global frame.

C. Incorporating Gait Constraints

We found empirically that the VINS on quadrupeds may accumulate significant drifts in the estimated trajectories, in particular, along the z-axis, primarily due to the contact impulse experienced in a walking robot. Although updating with leg measurements can help but certainly not eliminate the z-axis drift. This drift may cause the estimated position of the robot to deviate gradually from the actual ground plane where the robot is walking on, thus leading to serious issues for perception and planning. For example, due to this drift, the online ESDF map would be distorted, causing the previously explored ground areas to shift upwards and be misinterpreted as obstacles. Consequently, this erroneous mapping easily prevents the planner from recognizing obstacle-free paths on the ground plane. Regarding autonomous exploration, such misinterpretations can severely impact frontier-based exploration, as the inability to navigate through falsely labeled obstacles may hinder the robot to explore the environment effectively.

To address this issue, we propose to utilize the prior knowledge that the robot is primarily walking on a ground plane at a given time for the exploration scenarios under consideration (e.g., structured indoor environments); that is, we constrain the toe landmark point ${}^G \mathbf{p}_\tau$ on the ground plane so that the z-axis drift can be bounded at all times. Specifically, when the robot is standing, we extract a ground plane normal $\mathbf{n}_\pi$ using the four toe landmarks in global frame ${}^G \mathbf{p}_\tau$ as shown in Fig. 3. Every time the toe is in contact with the ground, and if the drift in z-axis with the ground plane exceeds a certain threshold, we further update the state with the *walk-on-plane constraint*, which can be obtained by projecting toe landmark on the ground plane $\mathbf{n}_\pi$ as follows:

$$\begin{aligned} z_\pi &:= 0 = \mathbf{n}_\pi^\top {}^G \mathbf{p}_\tau - 1 + \sigma_\pi \\ &= \mathbf{n}_\pi^\top ({}^I \mathbf{R}^\top {}^I \mathbf{p}_\tau(\boldsymbol{\theta}_m) + {}^G \mathbf{p}_I) - 1 + \sigma_\pi \end{aligned} \quad (10)$$

where ${}^I \mathbf{p}_\tau(\boldsymbol{\theta}_m)$ is the forward kinematics transformation from leg toe to the body frame $\{I\}$. We then perform update in our MSCKF-based VINS estimator with this walk-on-plane measurement whenever it is detected as valid.

¹Throughout this paper $\hat{\mathbf{x}}$ is used to denote the estimate of a random variable $\mathbf{x}$, while $\tilde{\mathbf{x}} = \mathbf{x} \ominus \hat{\mathbf{x}}$ is the error of this estimate. We define the orientation error quaternion, $\delta\theta$, as $\bar{q} \otimes \hat{q}^{-1} \simeq [\frac{1}{2}\delta\theta^\top \quad 1]^\top$ [39].

V. ONLINE MULTIMODAL ESDF MAPPING

The Online Mapping module fuses multimodal depth data from the D435i depth camera and VLP-16 (or OS1) LiDAR (see Fig. 1) to build an online high-resolution ESDF-based map to support planning.

A. Multimodal Depth Fusion

We integrate D435i depth camera and VLP-16 point cloud into the volumetric map by first using GPU-based ray tracing to identify and allocate visible voxel blocks. Each voxel center is transformed into the sensor frame and projected into the depth image to compute depth:

$$d = D \left[\pi_{\text{sensor}}(\mathbf{T}_{CL} \mathbf{p}_L) \right] \quad (11)$$

where π_{sensor} is the sensor-specific projection function (i.e., a pinhole model for cameras and a polar conversion for LiDAR), $\mathbf{T}_{CL}$ and $\mathbf{p}_L$ is the relative sensor pose and position to the layer. Depth values are then sampled, using nearest-neighbor for dense camera data or modified linear interpolation for sparse LiDAR scans, to update the map, efficiently fusing multi-modal data for real-time volumetric reconstruction. This fusion enhances the density and robustness of the depth measurements, particularly in low-texture environments. Notably, the LiDAR's wide field-of-view (FOV) and long-range capabilities make it well-suited for large-scale mapping, while the depth camera, oriented forward and slightly tilted towards the ground, refines the map in the immediate vicinity of the robot. This complementary sensing modalities enable accurate mapping of both distant structures and fine details of close distance.

B. ESDF Generation

Leveraging the off-the-shelf nvblox [12] which is a GPU-accelerated ESDF-mapping framework, we construct a TSDF whose value in a voxel ν is computed as a weighted average of measurements:

$$\phi(\nu) = \frac{\sum_i w_i \cdot \text{clip}(d_i - \|\nu - \mathbf{p}_i\|, -\epsilon, \epsilon)}{\sum_i w_i}, \quad (12)$$

where d_i is the depth measurement from the i -th sensor, $\mathbf{p}_i$ denotes the corresponding sensor position, ϵ is the truncation distance, and w_i reflects the confidence of the measurement. The clipping function limits the signed distance to the interval $[-\epsilon, \epsilon]$, thereby mitigating the influence of sensor noise and outliers. Subsequently, we derive the ESDF from the TSDF via an incremental wavefront expansion (see [15]). Specifically, when the TSDF value of a voxel changes, it is queued, and its neighbors are updated using the rule:

$$d(\nu_{\text{nb}}) \leftarrow \min(d(\nu_{\text{nb}}), d(\nu) + \|\nu_{\text{nb}} - \nu\|), \quad (13)$$

ensuring that the ESDF value of each voxel reflects the minimum Euclidean distance from the nearest obstacle. The sign of the ESDF is inherited from the TSDF, thereby distinguishing free space from occupied regions.

Note that accurate poses from the State Estimation module ensure that successive depth maps are accurately aligned within a common global frame, which is critical for maintaining an accurate map in dynamic environments. With real-time

updates enabled by GPU-based computation from nvblox, the resulting ESDF provides explicit distance-to-obstacle metrics that are essential for robust and efficient planning.

VI. EFFICIENT ESDF-BASED PLANNING

The objective of Local Planner is to generate a collision-free trajectory in the 2D configuration space $\mathbb{R}^2 \times SO(2)$ with configuration being $\mathbf{q} = [x \ y \ \theta]^\top$. The planner drives the robot from its initial state to a designated goal configuration under the following common assumptions:

- The robot operates on a plane parallel to XY -plane at a constant height z_{robot} , with altitude variations being negligible.
- The robot is modeled as a rigid cylinder with radius r_{robot} , mass m , and moment of inertia I about the Z -axis, ensuring collision detection, spatial reasoning, and consideration of its heavy body.
- The robot is equipped with a low-level locomotion controller that tracks commanded local velocities in XY -plane and yaw rate $\mathbf{u} = [u_x \ u_y \ u_\theta]^\top$, resulting in holonomic dynamics [43].

The equation of motion for the assumed dynamics, for which the motion is to be planned, is given by:

$$\ddot{\mathbf{q}} = \underbrace{\begin{bmatrix} -\frac{k}{m} & 0 & 0 \\ 0 & -\frac{k}{m} & 0 \\ 0 & 0 & -\frac{k_\theta}{I} \end{bmatrix}}_{\mathbf{A}} \dot{\mathbf{q}} + \underbrace{\begin{bmatrix} \frac{k}{m} \cos(\theta) & -\frac{k}{m} \sin(\theta) & 0 \\ \frac{k}{m} \sin(\theta) & \frac{k}{m} \cos(\theta) & 0 \\ 0 & 0 & \frac{k_\theta}{I} \end{bmatrix}}_{\mathbf{B}(\theta)} \mathbf{u}, \quad (14)$$

where k and k_θ are proportional constants characterizing the tracking performance of the low-level controller for planar linear and angular motion. This formulation describes a holonomic system, meaning the robot can move in any direction regardless of its orientation. A key advantage of this model is that, given a twice-differentiable configuration trajectory $\mathbf{q}(t)$, the control inputs can be computed as:

$$\mathbf{u}(t) = (\mathbf{B}(\theta(t)))^{-1} (\ddot{\mathbf{q}}(t) - \mathbf{A}\dot{\mathbf{q}}(t)). \quad (15)$$

We represent configuration space trajectories using Bézier curves due to their smoothness and intuitive shape control via control points. Moreover, the de Casteljau algorithm offers an efficient and numerically stable method for curve evaluation [44]. For a trajectory defined over the time interval $t \in [0, T]$, the Bézier curve is given by

$$\mathbf{q}(t) = \sum_{i=0}^n \binom{n}{i} \left(1 - \frac{t}{T}\right)^{n-i} \left(\frac{t}{T}\right)^i \mathbf{P}_i, \quad (16)$$

where $\mathbf{P}_i \in \mathbb{R}^3$ ($i = 1, \dots, n$) are the control points.

A. Problem Formulation

The objective local planner is to compute a trajectory that transitions the robot from its initial state $[\mathbf{q}_{\text{init}}, \dot{\mathbf{q}}_{\text{init}}]$ to a target configuration $\mathbf{q}_{\text{goal}}$ while avoiding obstacles. Although many methods for this aim exist—such as building safe corridors in a cluttered space [45]—they often require additional computational resources once a map is given. Instead, to avoid additional computation on the available map, inspired by [46], we *directly* leverage the already computed ESDF to formulate a constrained optimization problem. The cost and

constraints are evaluated on a trajectory discretized with N time interval $[0, T]$, that is, $q(t_k), t_k = k \frac{T}{N}$ for $k = 0, \dots, N$. The optimization problem is given by:

$$\begin{aligned}
\min_{\mathbf{P}} \quad & \sum_{k=0}^{N-1} \{w_1 \|\ddot{\mathbf{q}}(t_k)\|^2 + w_2 \|\dot{\mathbf{q}}(t_k)\|^2 - \\
& w_3 [x(t_k) \ y(t_k) \ z_{\text{robot}}]^\top\} + w_4 \|\mathbf{q}(T) \ominus \mathbf{q}_{\text{goal}}\|^2, \\
\text{s.t.} \quad & \mathbf{q}(0) = \mathbf{q}_{\text{init}}, \quad \dot{\mathbf{q}}(0) = \dot{\mathbf{q}}_{\text{init}}, \\
& \mathbf{u}_{\min} \leq \mathbf{u}(t_k) \leq \mathbf{u}_{\max}, \\
& d([x(t_k) \ y(t_k) \ z_{\text{robot}}]^\top) \geq d_{\text{safe}} + r_{\text{robot}}, \\
& \mathbf{q}(t_k) \text{ is given by (16).} \\
& \text{for } k = 0, \dots, N
\end{aligned} \tag{17}$$

In the above cost function: (i) The smoothness term $w_1 \|\ddot{\mathbf{q}}(t_k)\|^2$ and $w_2 \|\dot{\mathbf{q}}(t_k)\|^2$ penalizes accelerations and velocities. (ii) The clearance term $-w_3 d([x(t_k) \ y(t_k) \ z_{\text{robot}}]^\top)$ where $d(\cdot)$ is the ESDF (12) encourages trajectories that maintain a safe distance from obstacles. (iii) The goal-tracking term $w_4 \|\mathbf{q}(T) \ominus \mathbf{q}_{\text{goal}}\|^2$ softly enforces convergence to the target configuration. The initial state constraints are strictly enforced to the initial state of the robot. Bounds on input $\mathbf{u}_{\min} \leq \mathbf{u}(t_k) \leq \mathbf{u}_{\max}$ imposes a constraint on trajectory through the control law (15), while the collision avoidance constraint $d([x(t_k) \ y(t_k) \ z_{\text{robot}}]^\top) \geq d_{\text{safe}} + r_{\text{robot}}$ ensures a minimum distance from obstacles additional to model's dimensions.

B. Efficient Solution

The optimization problem is inherently nonlinear due to the nonlinearities in the ESDF for any nontrivial environment. We solve for the control points $\mathbf{P}_i$ using a gradient-descent method with line search, incorporating inequality constraints through penalty terms in the cost function. Gradients are computed via finite differences, and ESDF values are obtained via bilinear interpolation. The robot's initial state is strictly enforced by setting the first two control points as

$$\mathbf{P}_0 = \mathbf{q}_{\text{init}}, \quad \mathbf{P}_1 = \frac{\mathbf{q}_{\text{init}}}{n} + \dot{\mathbf{q}}_{\text{init}}$$

We then iteratively refine the remaining control points until the cost converges, after which we compute the control inputs using (15) that is commanded to the robot to execute the computed trajectory. Although penalty-based methods can tolerate trajectories that initially violate constraints, starting inside obstacles often causes divergence or increases the number of required iterations. To enhance solver stability, we generate an obstacle-free initial guess using the available ESDF, as outlined in Algorithm 1. While this heuristic does not guarantee absolute feasibility, in practice the solver can typically recover from any small infeasibilities and converge to a valid solution.

C. Experimental Validation

We tested our planner on the *Ghost Vision 60* quadrupedal robot, shown in Figure 1. This robot weighs around $m = 50\text{kg}$ and has a moment of inertia of $0.9\text{kg}\cdot\text{m}^2$ about its z -axis, and a maximum body length of 0.9m . For added safety,

Algorithm 1 Initial guess generation for local planner

Require: Initial configuration $\mathbf{q}_{\text{init}}$, Goal configuration $\mathbf{q}_{\text{goal}}$
Ensure: Discrete Trajectory $\{\mathbf{q}_0, \dots, \mathbf{q}_N\}$

- 1: Set $\mathbf{q}_0 \leftarrow$ Start position and orientation
- 2: Set time step $\Delta t = T/N$
- 3: **for** $i = 0$ to $N - 1$ **do**
- 4: $\mathbf{v}_k = \text{clip}(K(\mathbf{q}_{\text{goal}} - \mathbf{q}_0), \mathbf{v}_{\min}, \mathbf{v}_{\max})$
- 5: $\mathbf{q}_{\text{candidate}} = \mathbf{q}_k + \mathbf{v}_k \Delta t$
- 6: $\mathbf{q}_{k+1} \leftarrow \text{NearestFreePoint}(\mathbf{q}_{\text{candidate}}, d(\cdot))$
- 7: **end for**
- 8: $\mathbf{P}_{\text{init}} \leftarrow \text{FitBezier}(\{\mathbf{q}_0, \dots, \mathbf{q}_N\})$.

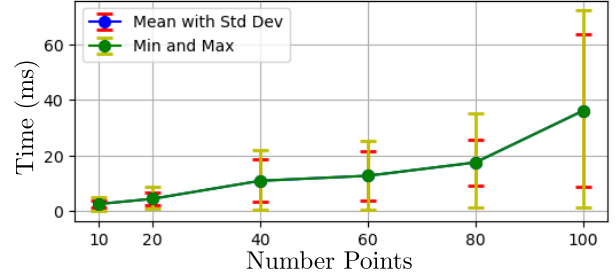


Fig. 4: Time required by the proposed local planner to find a solution for a trajectory of $T = 5\text{s}$ discretized into various number of points.

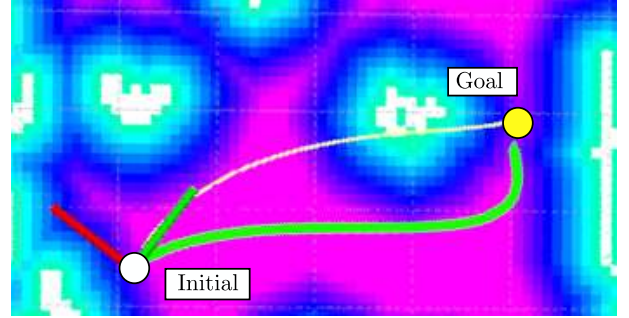


Fig. 5: A sample solution of the local planner. White curve from initial pose to goal pose is the initial guess generated with Algorithm 1. The thick green curve is the final optimized trajectory obtained from the local planner.

we set $d_{\text{safe}} + r_{\text{robot}} = 1\text{m}$. During walking, the robot's base remains at approximately 0.5m above the ground, so we extract an ESDF slice at $z_{\text{robot}} = 0.5\text{m}$. The proportional constants for the planning model (14) are selected to be $k = 150$ and $k_{\theta} = 7$ based on the observed step response of the robot to velocity commands. The maximum velocity commands set to $\mathbf{u}_{\max} = [0.5 \ 0.3 \ 0.5]^\top$ and $\mathbf{u}_{\min}$ of the same magnitude with negative sign.

In the optimization problem (17), cost and feasibility checks are evaluated only at N discrete time steps. Although increasing N yields more accurate collision checks, it also raises computational costs (see Fig. 4). Nevertheless, with a planning horizon of $T = 5\text{s}$ and $N = 100$ points, the solver finishes in under 70ms . For the rest of our experiments, we choose $N = 40$, corresponding to segments of about 7cm at the maximum speed—sufficiently safe for the robot's dimensions. Fig. 5 shows a sample trajectory plan obtained

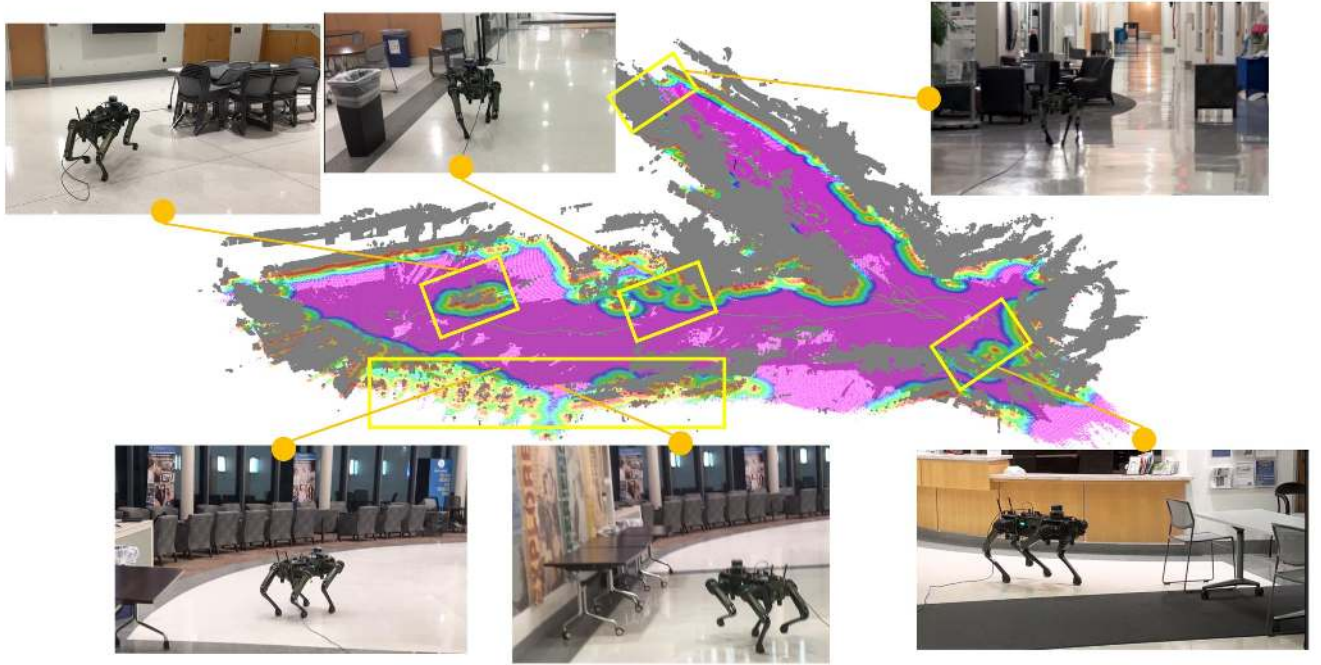


Fig. 6: Map overlaying ESDF (rainbow colors) and volumetric voxels (gray), generated online during autonomous exploration with our quadrupedal robot in an indoor environment, with a trajectory shown as a green line and the autonomy stack running on a laptop for visualization. The surrounding images show a third-person view of the robot exploring the environment corresponding to the location in the map.

for the Ghost Vision 60 robot in a cluttered space while the re-planning is being done at 15Hz.

VII. APPLICATION AND EXPERIMENT

With the autonomy stack presented in the preceding sections, we build an autonomous exploration application to verify the proposed autonomy capability, as autonomous exploration of an unknown environment is one of the fundamental problems in robotics with great potential in practice. One of the most successful and foundational approach for this problem is of the Frontier-based exploration [47].

Specifically, this approach relies on identifying frontiers, which are the boundary regions separating the explored area and unexplored territory. The ESDF map generated by the online mapping module is converted to occupancy map at a inflated level to consider robot's dimension and additional clearance $d_{\text{safe}} + r_{\text{robot}}$, which is then searched to identify the frontier regions. These frontier regions are clustered based on their proximity to the robot's current position and their size, with centroids computed for each cluster. A global planner employs the Jump Point Search (JPS) [48] algorithm to compute paths from the robot's position to each frontier centroid, evaluating their visibility and accessibility. To optimize the exploration efficiency, the Traveling Salesman Problem (TSP) framework is employed to determine an optimal sequence for visiting the frontier

centroids. A waypoint is generated at a look-away distance within a specified range from the robot position by following the path to the first frontier centroid in the queue. This exploration step is repeated each time the ESDF map is updated by the online mapping module and a waypoint is generated. This waypoint along with its heading from the robot's position is used by the local planner to generate feasible body velocity commands and an efficient trajectory.

Clearly, this exploration is essentially enabled by the state estimation, online mapping and local planning. As a proof-of-concept test, we deploy this autonomous exploration system onto our quadruped and perform a preliminary experiment in a GPS-denied indoor environment detailed next.

A. Preliminary Testing

We evaluated the baseline system of autonomous exploration on *Ghost Vision 60* in the STAR Health Sciences Complex at the University of Delaware. The third-person view of the robot exploring the environment in the location of the generated map are shown in Fig. 6. As evident, some parts of the environment has varying lighting conditions and a shiny floor, making VINS feature detection and tracking challenging. Despite that, the proposed kinematics-constrained, gait-constrained VINS-based state estimator was able to provide accurate localization to fundamentally support the exploration. Especially, we further evaluated separately the

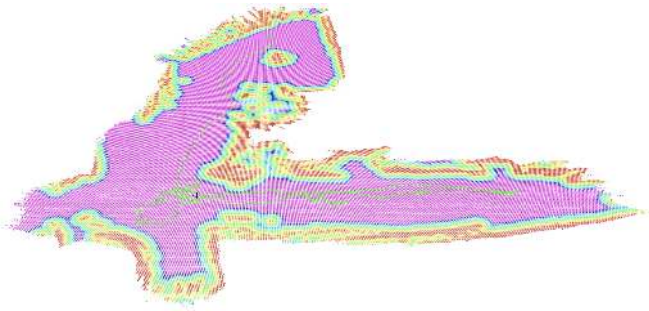


Fig. 7: The ESDF map generated online during exploration.

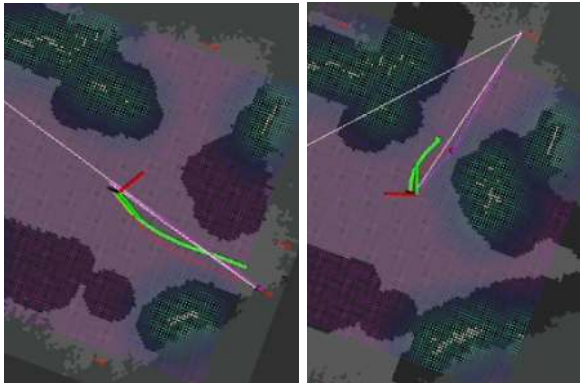


Fig. 8: Snapshots of local planner behavior during the test. White represents the frontier visiting order. And arrows at the frontiers. The global plan is represented in violato.

proposed state estimation module in a start-end loop while manually controlling the robot. The total trajectory length is 160.1m, with a drift of 2.07m, accounting for 1.29% of the total trajectory. The Z-axis drift is 0.5cm and it is consistently bounded to 1 ± 0.5 cm.

The online ESDF mapping uses the state estimates and fuses LiDAR and depth data to generate a refined maps as shown in Fig. 7. The frontier exploration module processes the ESDF map to identify frontier regions, clusters them, and computes the centroids of each cluster. The global planner then generates a path connecting all the frontier centroids shown in white line in Fig. 8 and gives a waypoint to the local planner which considers the kinodynamic constraints and generates a trajectory shown in green. These snapshots visualize the real-time behavior of our planner and confirm our expectation.

VIII. CONCLUSIONS AND FUTURE WORK

Quadrupedal robots are gaining traction and offer significant potential for autonomous applications. In this paper, we have taken up this challenge by developing an efficient quadrupedal autonomy stack that integrates multimodal state estimation, online mapping, and ESDF-based planning, going beyond just locomotion. In particular, we proposed a lightweight, kinematics-constrained, gait-informed visual-inertial estimation module. Leveraging precise localization, we developed an online, multimodal ESDF-based mapping framework that efficiently fuses LiDAR and RGBD data, and implemented efficient ESDF-based planning for resource-constrained environments. Finally, we integrated these autonomy algorithms with the *Ghost Vision 60* quadrupedal robot

and preliminarily validated the system through autonomous exploration in a GPS-denied indoor environment. In the future, we will further optimize our modular algorithms for estimation, perception, and planning. Expanding on our quadrupedal autonomy stack, we will develop sample return or data collection applications while testing in more challenging environments.

REFERENCES

- [1] S. Ha, J. Lee, M. van de Panne, Z. Xie, W. Yu, and M. Khadiv, "Learning-based legged locomotion: State of the art and future perspectives," *International Journal of Robotics Research*, 2025.
- [2] A. Hamrani, M. M. Rayhan, T. Mackenson, D. McDaniel, and L. Lagos, "Smart quadruped robotics: a systematic review of design, control, sensing and perception," *Advanced Robotics*, vol. 39, no. 1, pp. 3–29, 2024.
- [3] C. Gehring, C. Dario Bellicoso, P. Fankhauser, S. Coros, and M. Hutter, "Quadrupedal locomotion using trajectory optimization and hierarchical whole body control," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 4788–4794.
- [4] S. Fahmi, G. Fink, and C. Semini, "On state estimation for legged locomotion over soft terrain," *IEEE Sensors Letters*, vol. 5, no. 1, pp. 1–4, 2021.
- [5] G. Ji, J. Mun, H. Kim, and J. Hwangbo, "Concurrent training of a control policy and a state estimator for dynamic and robust legged locomotion," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4630–4637, 2022.
- [6] M. Camurri, M. Ramezani, S. Nobili, and M. Fallon, "Pronto: A multi-sensor state estimator for legged robots in real-world scenarios," *Frontiers in Robotics and AI*, vol. 7, p. 68, 2020.
- [7] D. Wisth, M. Camurri, and M. Fallon, "Vilens: Visual, inertial, lidar, and leg odometry for all-terrain legged robots," *IEEE Transactions on Robotics*, vol. 39, no. 1, pp. 309–326, 2022.
- [8] R. Hartley, M. Ghaffari, R. M. Eustice, and J. W. Grizzle, "Contact-aided invariant extended kalman filtering for robot state estimation," *The International Journal of Robotics Research*, vol. 39, no. 4, pp. 402–430, 2020.
- [9] L. Han, F. Gao, B. Zhou, and S. Shen, "Fiesta: Fast incremental euclidean distance fields for online motion planning of aerial robots," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 4423–4430.
- [10] T. Corbères, C. Mastalli, W. Merkt, I. Havoutis, M. Fallon, N. Mansard, T. Flayols, S. Vijayakumar, and S. Tonneau, "Perceptive locomotion through whole-body mpc and optimal region selection," *arXiv preprint arXiv:2305.08926*, 2023.
- [11] R. Grandia, F. Jenelten, S. Yang, F. Farshidian, and M. Hutter, "Perceptive locomotion through nonlinear model-predictive control," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3402–3421, 2023.
- [12] A. Millane, H. Oleynikova, E. Wirbel, R. Steiner, V. Ramasamy, D. Tingdahl, and R. Siegwart, "nvblox: Gpu-accelerated incremental signed distance field mapping," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024, pp. 2698–2705.
- [13] Y. Fan, Z. Pei, C. Wang, M. Li, Z. Tang, and Q. Liu, "A review of quadruped robots: Structure, control, and autonomous motion," *Advanced Intelligent Systems*, vol. 6, no. 6, p. 2300783, 2024.
- [14] B. Curless and M. Levoy, "A volumetric method for building complex models from range images," in *Proceedings of the 23rd annual conference on Computer graphics and interactive techniques*, 1996, pp. 303–312.
- [15] H. Oleynikova, Z. Taylor, M. Fehr, R. Siegwart, and J. Nieto, "Voxblox: Incremental 3d euclidean signed distance fields for on-board mav planning," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vancouver, Canada, Sep. 2017, pp. 1366–1373.
- [16] G. Erni, J. Frey, T. Miki, M. Mattamala, and M. Hutter, "Mem: Multimodal elevation mapping for robotics and learning," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 11 011–11 018.
- [17] G. A. Kumar, J. H. Lee, J. Hwang, J. Park, S. H. Youn, and S. Kwon, "Lidar and camera fusion approach for object distance estimation in self-driving vehicles," *Symmetry*, vol. 12, no. 2, p. 324, 2020.

- [18] J. S. Berrio, M. Shan, S. Worrall, and E. Nebot, "Camera-lidar integration: Probabilistic sensor fusion for semantic mapping," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 7637–7652, 2021.
- [19] M. Bloesch, M. Hutter, C. Gehring, M. A. Hoepflinger, and R. Siegwart, "Kinematic batch calibration for legged robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 2542–2547.
- [20] M. Bloesch, M. Burri, H. Sommer, R. Siegwart, and M. Hutter, "The two-state implicit filter recursive estimation for mobile robots," *IEEE Robotics and Automation Letters*, vol. 3, no. 1, pp. 573–580, 2017.
- [21] M. Camurri, M. Fallon, S. Bazeille, A. Radulescu, V. Barasuol, D. G. Caldwell, and C. Semini, "Probabilistic contact estimation and impact detection for state estimation of quadruped robots," *IEEE Robotics and Automation Letters (RA-L)*, vol. 2, no. 2, pp. 1023–1030, 2017.
- [22] F. Jenelten, J. Hwangbo, F. Tresoldi, C. D. Bellicoso, and M. Hutter, "Dynamic locomotion on slippery ground," *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, no. 4, pp. 4170–4176, 2019.
- [23] M. Bloesch, M. Hutter, M. A. Hoepflinger, S. Leutenegger, C. Gehring, C. D. Remy, and R. Siegwart, "State estimation for legged robots-consistent fusion of leg kinematics and imu," *Robotics*, 2013.
- [24] D. Wisth, M. Camurri, and M. Fallon, "Robust legged robot state estimation using factor graph optimization," *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, no. 4, pp. 4507–4514, 2019.
- [25] R. Hartley, M. G. Jadidi, L. Gan, J.-K. Huang, J. W. Grizzle, and R. M. Eustice, "Hybrid contact preintegration for visual-inertial-contact state estimation using factor graphs," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 3783–3790.
- [26] Y. Kim, B. Yu, E. M. Lee, J.-h. Kim, H.-w. Park, and H. Myung, "Step: State estimator for legged robots using a preintegrated foot velocity factor," *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 4456–4463, 2022.
- [27] C. Burgul, W. Lee, P. Geneva, and G. Huang, "Online determination of legged kinematics," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 9043–9049.
- [28] S. Yang, Z. Zhang, Z. Fu, and Z. Manchester, "Cerberus: Low-drift visual-inertial-leg odometry for agile locomotion," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 4193–4199.
- [29] F. Jenelten, T. Miki, A. E. Vijayan, M. Bjelonic, and M. Hutter, "Perceptive locomotion in rough terrain – online foothold optimization," *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5370–5376, 2020.
- [30] D. Kim, D. Carballo, J. Di Carlo, B. Katz, G. Bledt, B. Lim, and S. Kim, "Vision aided dynamic exploration of unstructured terrain with a small-scale quadruped robot," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 2464–2470.
- [31] T. Dudzik, M. Chignoli, G. Bledt, B. Lim, A. Miller, D. Kim, and S. Kim, "Robust autonomous navigation of a small-scale quadruped robot in real-world environments," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 3664–3671.
- [32] C. Mastalli, I. Havoutis, M. Focchi, D. G. Caldwell, and C. Semini, "Motion planning for quadrupedal locomotion: Coupled planning, terrain mapping, and whole-body control," *IEEE Transactions on Robotics*, vol. 36, no. 6, pp. 1635–1648, 2020.
- [33] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.
- [34] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 3803–3810.
- [35] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "Rma: Rapid motor adaptation for legged robots," 2021.
- [36] A. Kumar, Z. Li, J. Zeng, D. Pathak, K. Sreenath, and J. Malik, "Adapting rapid motor adaptation for bipedal robots," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 1161–1168.
- [37] C. S. Imai, M. Zhang, Y. Zhang, M. Kierebiński, R. Yang, Y. Qin, and X. Wang, "Vision-guided quadrupedal locomotion in the wild with multi-modal delay randomization," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 5556–5563.
- [38] Z. Fu, A. Kumar, A. Agarwal, H. Qi, J. Malik, and D. Pathak, "Coupling vision and proprioception for navigation of legged robots," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 17 252–17 262.
- [39] N. Trawny and S. I. Roumeliotis, "Indirect Kalman filter for 3D attitude estimation," University of Minnesota, Dept. of Comp. Sci. & Eng., Tech. Rep., Mar. 2005.
- [40] Y. Yang, P. Geneva, X. Zuo, and G. Huang, "Online self-calibration for visual-inertial navigation: Models, analysis, and degeneracy," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3479–3498, 2023.
- [41] A. I. Mourikis and S. I. Roumeliotis, "A multi-state constraint Kalman filter for vision-aided inertial navigation," in *Proceedings of the IEEE International Conference on Robotics and Automation*, Rome, Italy, 2007, pp. 3565–3572.
- [42] P. Geneva, K. Eickenhoff, W. Lee, Y. Yang, and G. Huang, "OpenVINS: a research platform for visual-inertial estimation," in *Proc. of the IEEE International Conference on Robotics and Automation*, Paris, France, 2020, pp. 4666–4672. [Online]. Available: https://github.com/rpng/open_vins
- [43] J.-K. Huang and J. W. Grizzle, "Efficient anytime clf reactive planning system for a bipedal robot on undulating terrain," *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 2093–2110, 2023.
- [44] J.-w. Choi, R. Curry, and G. Elkaim, "Path planning based on bézier curve for autonomous ground vehicles," in *Advances in Electrical and Electronics Engineering - IAENG Special Edition of the World Congress on Engineering and Computer Science 2008*, 2008, pp. 158–166.
- [45] K. S. Narkhede, A. M. Kulkarni, D. A. Thanki, and I. Poulakakis, "A sequential mpc approach to reactive planning for bipedal robots using safe corridors in highly cluttered environments," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 11 831–11 838, 2022.
- [46] H. Oleynikova, M. Burri, Z. Taylor, J. Nieto, R. Siegwart, and E. Galceran, "Continuous-time trajectory optimization for online uav replanning," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 5332–5339.
- [47] B. Yamauchi, "A frontier-based approach for autonomous exploration," in *Proceedings 1997 IEEE International Symposium on Computational Intelligence in Robotics and Automation CIRA'97. 'Towards New Computational Principles for Robotics and Automation'*, 1997, pp. 146–151.
- [48] D. Harabor and A. Grastien, "Online graph pruning for pathfinding on grid maps," in *Proceedings of the Twenty-Fifth AAAI Conference on Artificial Intelligence*, ser. AAAI'11. AAAI Press, 2011, p. 1114–1119.